Bits, Bytes and Integers

COMP402127: Introduction to Computer Systems

https://xjtu-ics.github.io/

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Today: Bits, Bytes, and Integers

Representing information as bits

Bit-level manipulations

Integers

- Representation: unsigned and signed
- Conversion, casting
- Expanding, truncating
- Addition, multiplication, shifting

Representations in memory, pointers, strings

Everything is bits

Each bit is 0 or 1

By encoding/interpreting sets of bits in various ways

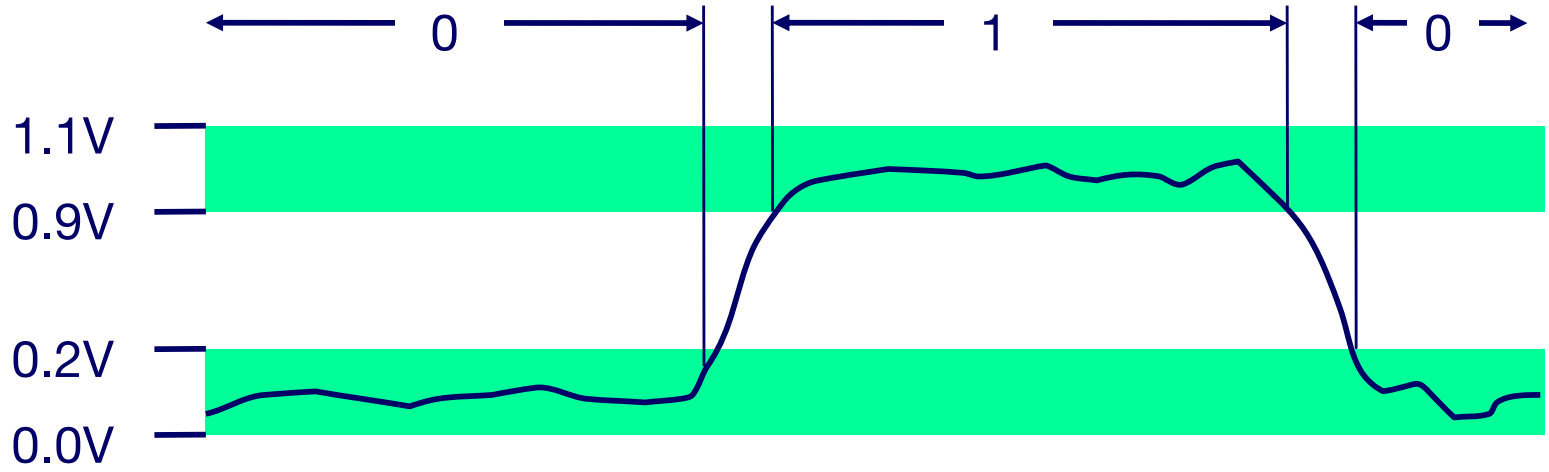
Computers determine what to do (instructions)

… and represent and manipulate numbers, sets, strings, etc…

Why bits? Electronic Implementation

Easy to store with bistable elements (双稳态器件)

Reliably transmitted on noisy and inaccurate wires



For example, can count in binary

Base 2 Number Representation

Represent 985_{10} as 1111011001_{2}

Represent 1.20_{10} as $1.0011001100110011[0011]\ldots_{2}$

Represent 1.5213×10^{4} as $1.1101101101101_{2} \times 2^{13}$

Encoding Byte Values

Byte = 8 bits

- Binary 00000000_2 to 11111111_2
- Decimal: 0_{10} to 255_{10}
- Hexadecimal 00_{16} to FF_{16}
  - Base 16 number representation
  - Use characters ‘0’ to ‘9’ and ‘A’ to ‘F’
  - Write $FA1D37B_{16}$ in C as
    - 0xFA1D37B
    - 0xfa1d37b

Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

985: 0011 1101 1001

3 D 9

这个世界上有10种人

一种懂二进制

还有一种不懂三进制并且以为这是二进制的

还有一种不懂四进制并且以为这是三进制的

……

Example Data Representations

C Data Type	Typical 32-bit	Typical 64-bit
`char`	1	1
`short`	2	2
`int`	4	4
`long`	4	8
`float`	4	4
`double`	8	8
pointer	4	8

Example Data Representations

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`float`	4	4
`double`	8	8
pointer	4	8
	“ILP32”	“LP64”

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Boolean Algebra

Developed by George Boole in 19th Century

Algebraic representation of logic

Encode “True” as 1 and “False” as 0

And

A&B = 1 when both A=1 and B=1

&	0	1
0	0	0
1	0	1

Or

A|B = 1 when either A=1 or B=1 or both

|	0	1
0	0	1
1	1	1

Not

~A = 1 when A=0

~	0	1
	1	0

Exclusive-Or (Xor)

A^B = 1 when A=1 or B=1, but not both

^	0	1
0	0	1
1	1	0

General Boolean Algebras

Operate on Bit Vectors

Operations applied bitwise

```
  01101001     01101001     01101001
& 01010101   | 01010101   ^ 01010101   ~ 01010101
  01000001     01111101     00111100     10101010
```

Bit-Level Operations in C

Operations `&`, `|`, `~`, `^` Available in C

Apply to any "integral" data type

`long, int, short, char, unsigned`

View arguments as bit vectors

Arguments applied bit-wise

Contrast: Logic Operations in C

Contrast to Bit-Level Operators

Logic Operations: &&, ||, !

- View 0 as "False"
- Anything nonzero as "True"
- Always return 0 or 1
- Early termination

Examples (char data type)

```
!0x41 →  0x00
!0x00 →  0x01
!!0x41→  0x01

0x69 && 0x55 →  0x01
0x69 || 0x55 →  0x01
p && *p
```

(avoids null pointer access)

Watch out for && vs. & (and || vs. |)…
Super common C programming pitfall!

Shift Operations

Left Shift: `x << y`

Shift bit-vector **`x`** left **`y`** positions

- Throw away extra bits on left

Fill with 0's on right

Right Shift: `x >> y`

Shift bit-vector **`x`** right **`y`** positions

Throw away extra bits on right

Logical shift

Fill with 0's on left

Arithmetic shift

Replicate most significant bit on left

Undefined Behavior

Shift amount < 0 or ≥ word size

Argument **`x`**	`01100010`
`<< 3`	`00010000`
Log. `>> 2`	`00011000`
Arith. `>> 2`	`00011000`

Argument **`x`**	`10100010`
`<< 3`	`00010000`
Log. `>> 2`	`00101000`
Arith. `>> 2`	`11101000`

Today: Bits, Bytes, and Integers

Representing information as bits

Bit-level manipulations

Integers

- Representation: unsigned and signed, negation
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Representations in memory, pointers, strings

Question?

```
int foo = -1;
unsigned bar = 1;

(foo < bar) == true ?
```

Encoding "Integers"

Unsigned

Given a bit vector x, w bits long…

$$\mathrm{B2U}(x) = \sum_{i=0}^{w-1} x_i \cdot 2^i$$

Signed (twos complement)

$$\mathrm{B2T}(x) = -x_{w-1} \cdot 2^{w-1} + \sum_{i=0}^{w-2} x_i \cdot 2^i$$

Sign Bit

Examples (w = 5)

±16	8	4	2	1
0	1	0	1	0

$$0 + 8 + 0 + 2 + 0 = 10$$

16	8	4	2	1
1	0	1	1	0
-16	8	4	2	1

$$16 + 0 + 4 + 2 + 0 = 22$$

$$-16 + 0 + 4 + 2 + 0 = -10$$

Negation: Complement & Increment

Negate through complement and increase

```
~x + 1 == -x
```

Why?

```
-x + x      == 0  (by definition)
~x + x      == 1111…111 == -1
~x + x + 1  == 0
(~x+1) + x  == 0
 ~x+1       == -x
```

x	1	0	0	1	1	1	0	1
+ ~x	0	1	1	0	0	0	1	0
−1	1	1	1	1	1	1	1	1

Example: x = 985

	Decimal	Hex	Binary
x	985	03D9	0000 0011 1101 1001
~x	-986	FC26	1111 1100 0010 0110
~x+1	-985	FC27	1111 1100 0010 0111
y	-985	FC27	1111 1100 0010 0111

Complement & Increment Examples

$x = 0$

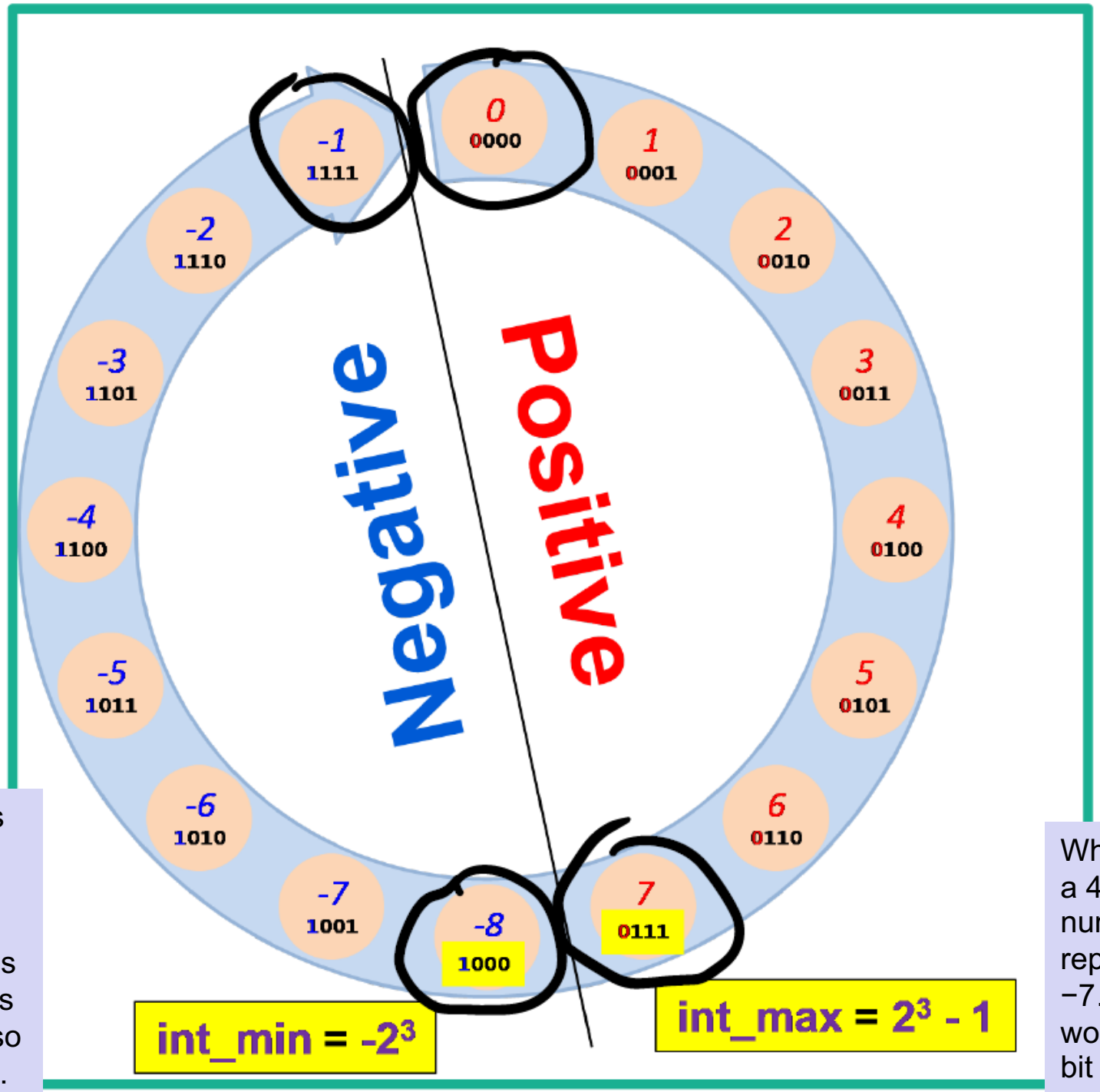
	Decimal	Hex	Binary
0	0	00 00	00000000 00000000
~0	-1	FF FF	11111111 11111111
~0+1	0	00 00	00000000 00000000

$x = T_{\min}$

	Decimal	Hex	Binary
x	-32768	80 00	10000000 00000000
~x	32767	7F FF	01111111 11111111
~x+1	-32768	80 00	10000000 00000000

Oops!
It's still
negative!

Eight negative values:
$-1, -2, \ldots, -8$



Eight *non*-negative values:
$0, 1, \ldots, 7$

Mathematicians would prefer it if a 4-bit signed number could represent values $-8 \ldots 8$, but that's $2^4 + 1$ values, so they won't all fit.

What if we made a 4-bit signed number only represent values $-7 \ldots 7$? Then we wouldn't be using bit pattern `1000`…

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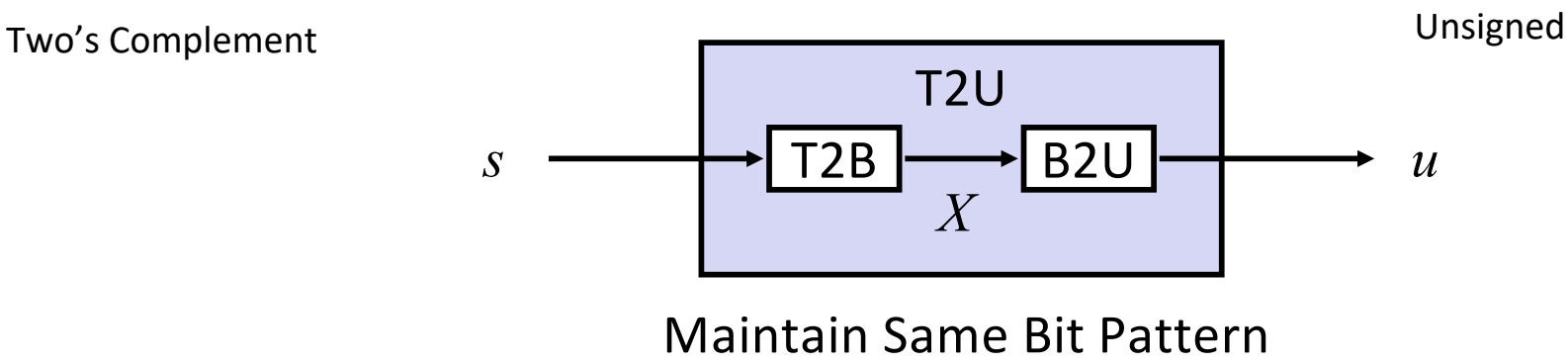
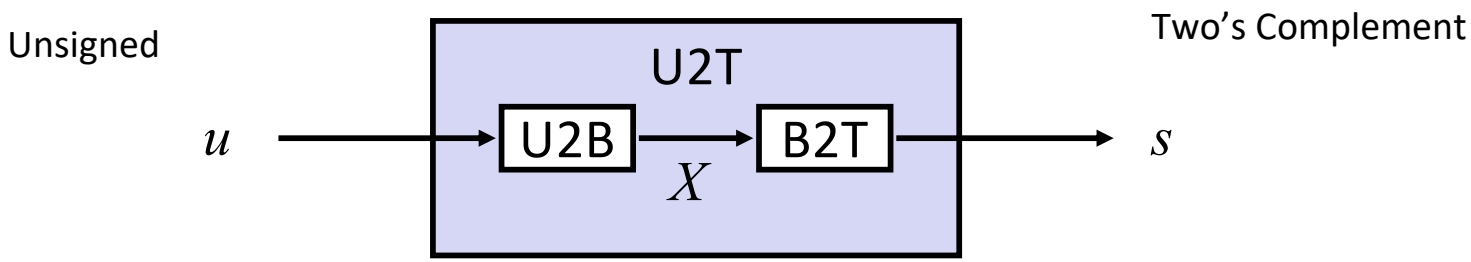
Mapping Between Signed & Unsigned



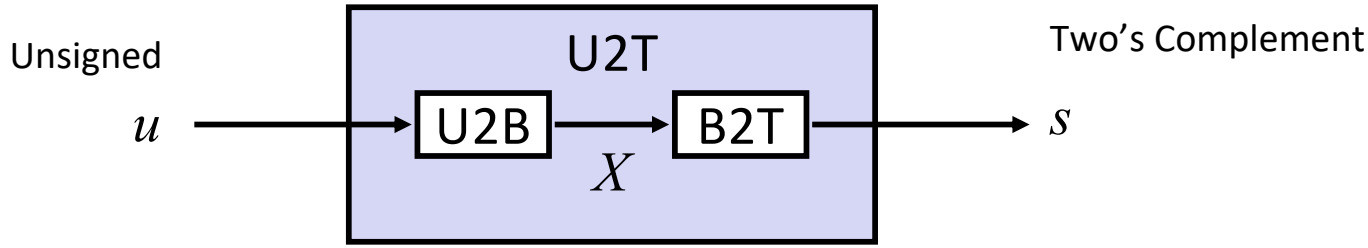
Maintain Same Bit Pattern



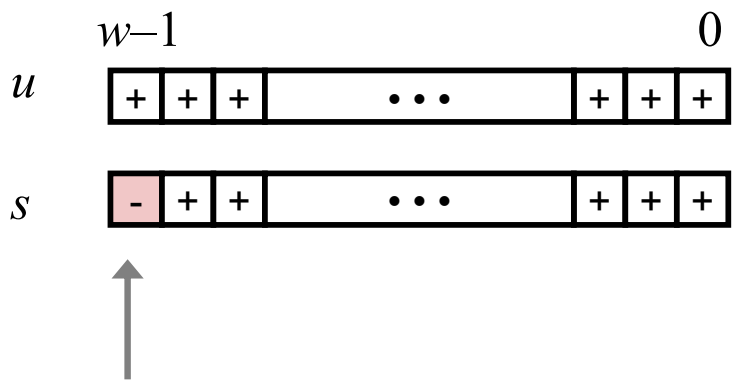
Maintain Same Bit Pattern

Mappings between unsigned and two's complement numbers:
Keep bit representations and reinterpret

Relation between Signed & Unsigned



Maintain Same Bit Pattern



Large positive weight
becomes
Large negative weight

Mapping Signed ↔ Unsigned

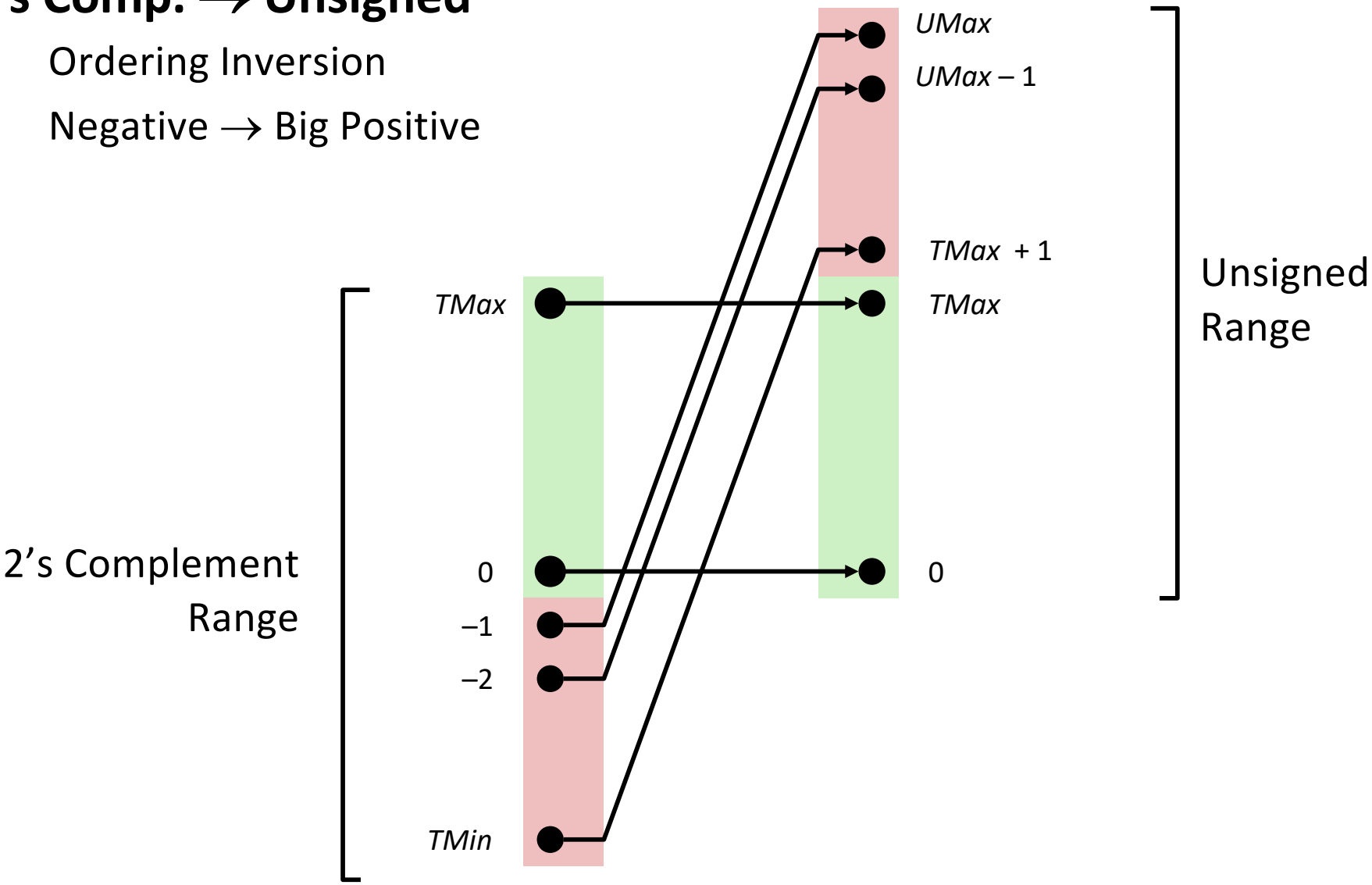
Bits	Signed		Unsigned
0000	0		0
0001	1		1
0010	2		2
0011	3	=	3
0100	4		4
0101	5		5
0110	6		6
0111	7		7
1000	-8		8
1001	-7		9
1010	-6		10
1011	-5	± 16	11
1100	-4		12
1101	-3		13
1110	-2		14
1111	-1		15

Conversion Visualized

2's Comp. → Unsigned

Ordering Inversion

Negative → Big Positive



Signed vs. Unsigned in C

Constants

By default are considered to be signed integers

Unsigned if have “U” as suffix

```
0U, 4294967259U
```

Casting

Explicit casting between signed & unsigned same as U2T and T2U

```
int tx, ty;
unsigned ux, uy;
tx = (int) ux;
uy = (unsigned) ty;
```

Implicit casting also occurs via assignments and procedure calls

```
tx = ux;
uy = ty;
```

```
int fun(unsigned u);
uy = fun(tx);
```

Question?

example02.c

```
int foo = -1;
unsigned bar = 1;

foo < bar == true ?
```

Casting Surprises

Expression Evaluation

If there is a mix of unsigned and signed in single expression,

signed values implicitly cast to unsigned

Including comparison operations **<, >, ==, <=, >=**

Examples:

Constant 1	Constant 2	Relation	Evaluation
`0`	`0U`	==	Unsigned
`-1`	`0`	<	Signed
`-1`	`0U`	>	Unsigned
`INT_MAX`	`INT_MIN`	>	Signed
`(unsigned)INT_MAX`	`INT_MIN`	<	Unsigned
`-1`	`-2`	>	Signed
`(unsigned)-1`	`-2`	>	Unsigned
`INT_MAX`	`((unsigned)INT_MAX) + 1`	<	Unsigned
`INT_MAX`	`(int)(((unsigned)INT_MAX) + 1)`	>	Signed

Summary
Casting Signed ↔ Unsigned: Basic Rules

Bit pattern is maintained

But reinterpreted

Can have unexpected effects: adding or subtracting 2^w

Expression containing signed and unsigned int

`int` is cast to `unsigned`!!

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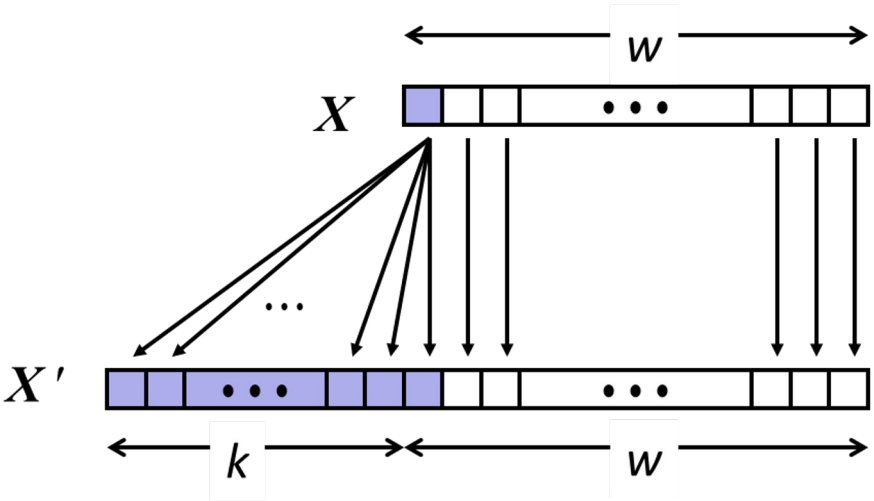
Question?

example03.c

```
int x = 0x8000;
short sx = (short) x;
int y = sx;
```

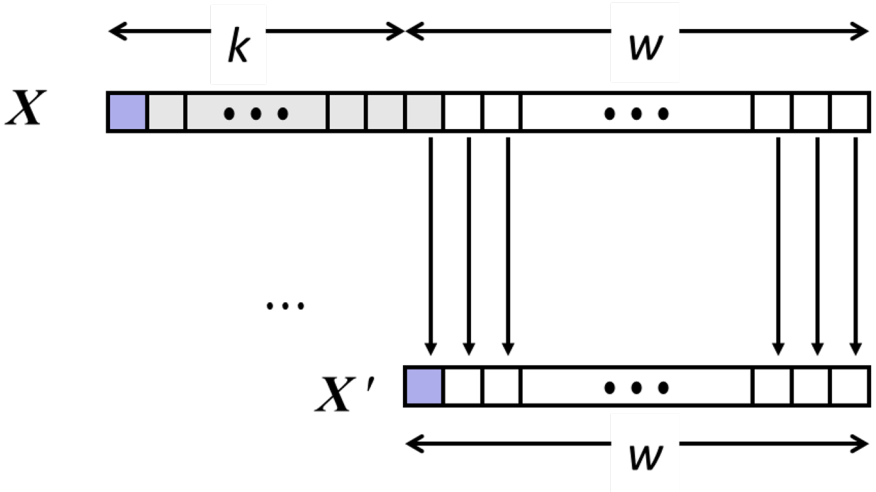
Sign Extension and Truncation

Sign Extension



Make ***K*** copies of sign bit

Truncation



Chop off ***K*** highest bits

Sign Extension: Simple Example

Positive number

		-16	8	4	2	1
10 =		0	1	0	1	0

	-32	16	8	4	2	1
10 =	0	0	1	0	1	0

Negative number

		-16	8	4	2	1
-10 =		1	0	1	1	0

	-32	16	8	4	2	1
-10 =	1	1	0	1	1	0

Truncation: Simple Example

No sign change

	-16	8	4	2	1
2 =	0	0	0	1	0

	-8	4	2	1
2 =	0	0	1	0

	-16	8	4	2	1
-6 =	1	1	0	1	0

	-8	4	2	1
-6 =	1	0	1	0

Sign change

	-16	8	4	2	1
10 =	0	1	0	1	0

	-8	4	2	1
-6 =	1	0	1	0

	-16	8	4	2	1
-10 =	1	0	1	1	0

	-8	4	2	1
6 =	0	1	1	0

Question?

example03.c

```
int x = 0x8000;
short sx = (short) x;
int y = sx;
```

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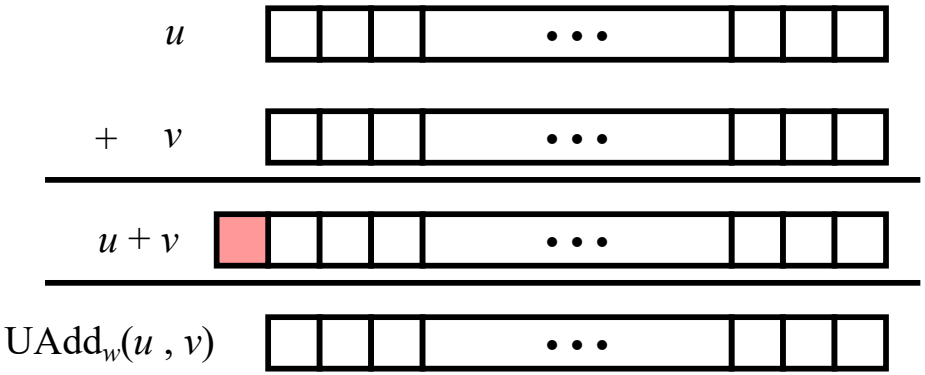
Unsigned Addition

Operands: *w* bits

True Sum: *w*+1 bits

Discard Carry: *w* bits



Standard Addition Function

Ignores carry output

```
unsigned char     1110 1001      E9      233
                + 1101 0101    + D5    + 213
                ___________    ____    _____
                ___________    ____    _____
```

Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

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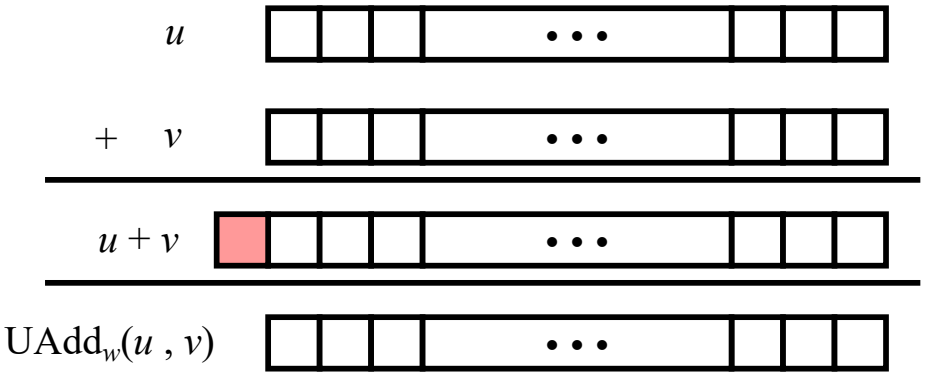


Standard Addition Function

Ignores carry output

```
unsigned char     1110 1001      E9      233
                +  1101 0101   + D5    + 213
                 1 1011 1110    1BE      446
                   1011 1110     BE      190
```

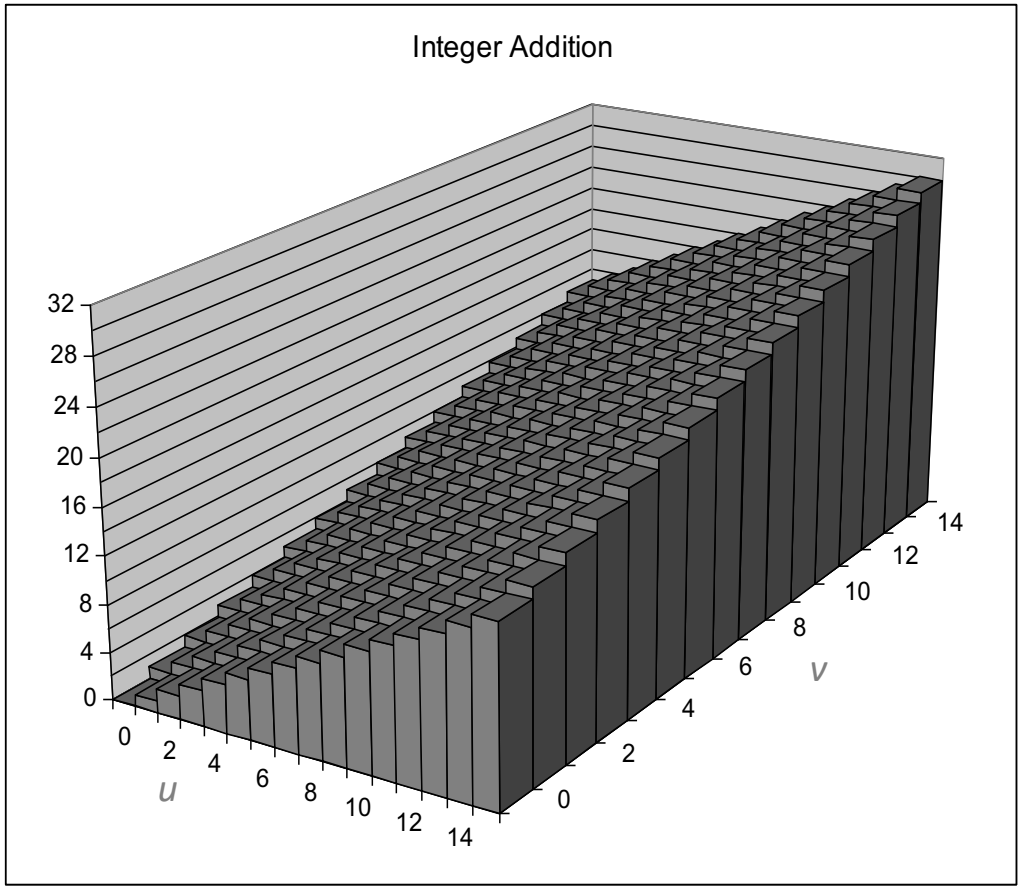
Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

Visualizing (Mathematical) Integer Addition

Integer Addition

- 4-bit integers *u*, *v*
- Compute true sum $\text{Add}_4(u\ ,\ v)$
- Values increase linearly with *u* and *v*
- Forms planar surface

$\text{Add}_4(u\ ,\ v)$

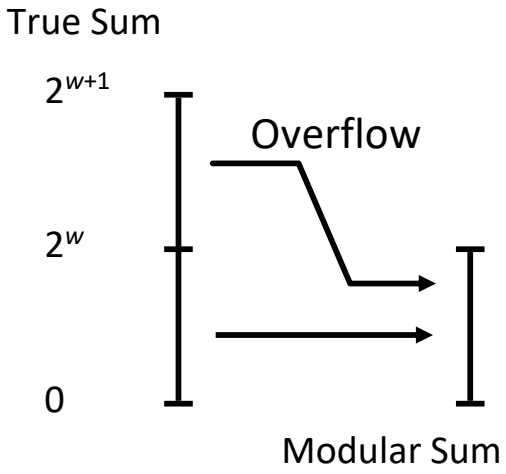


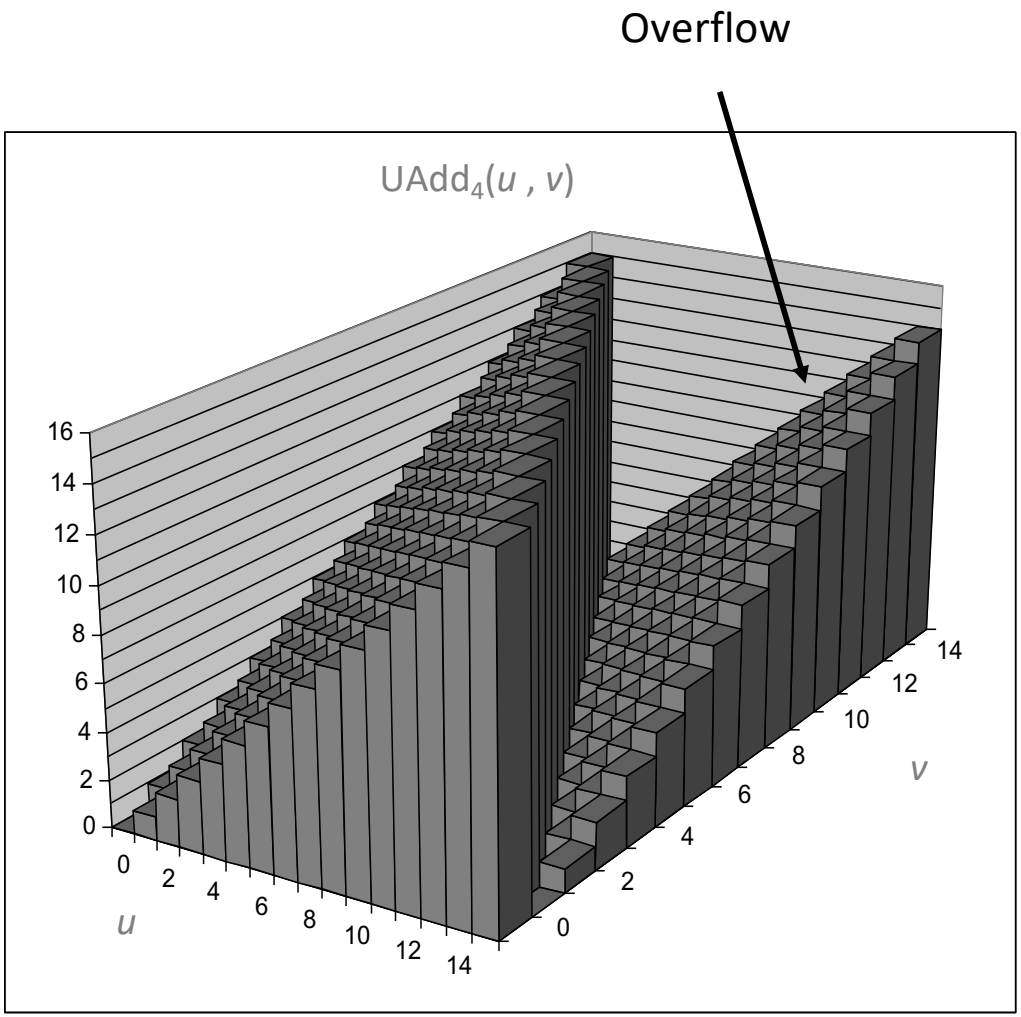
Visualizing Unsigned Addition

Wraps Around

If true sum ≥ 2^w

At most once



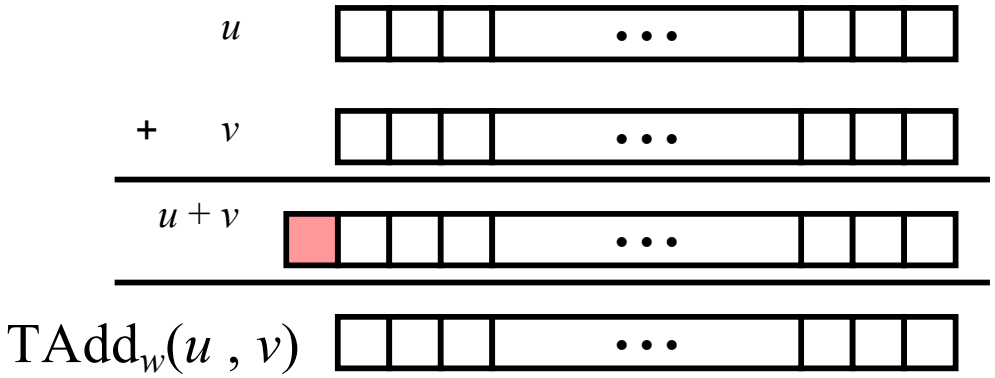


Two's Complement Addition

Operands: *w* bits

True Sum: *w*+1 bits

Discard Carry: *w* bits



TAdd and UAdd have Identical Bit-Level Behavior

Signed vs. unsigned addition in C:

```
int s, t, u, v;
s = (int) ((unsigned) u + (unsigned) v);
t = u + v
```

Will give `s == t`

	1110 1001	E9	-23
+	1101 0101	+ D5	+ -43
	1 1011 1110	1BE	-66
	1011 1110	BE	-66

Visualizing 2's Complement Addition

Values

4-bit two's comp.

Range from -8 to +7

Wraps Around

If sum $\geq 2^{w-1}$

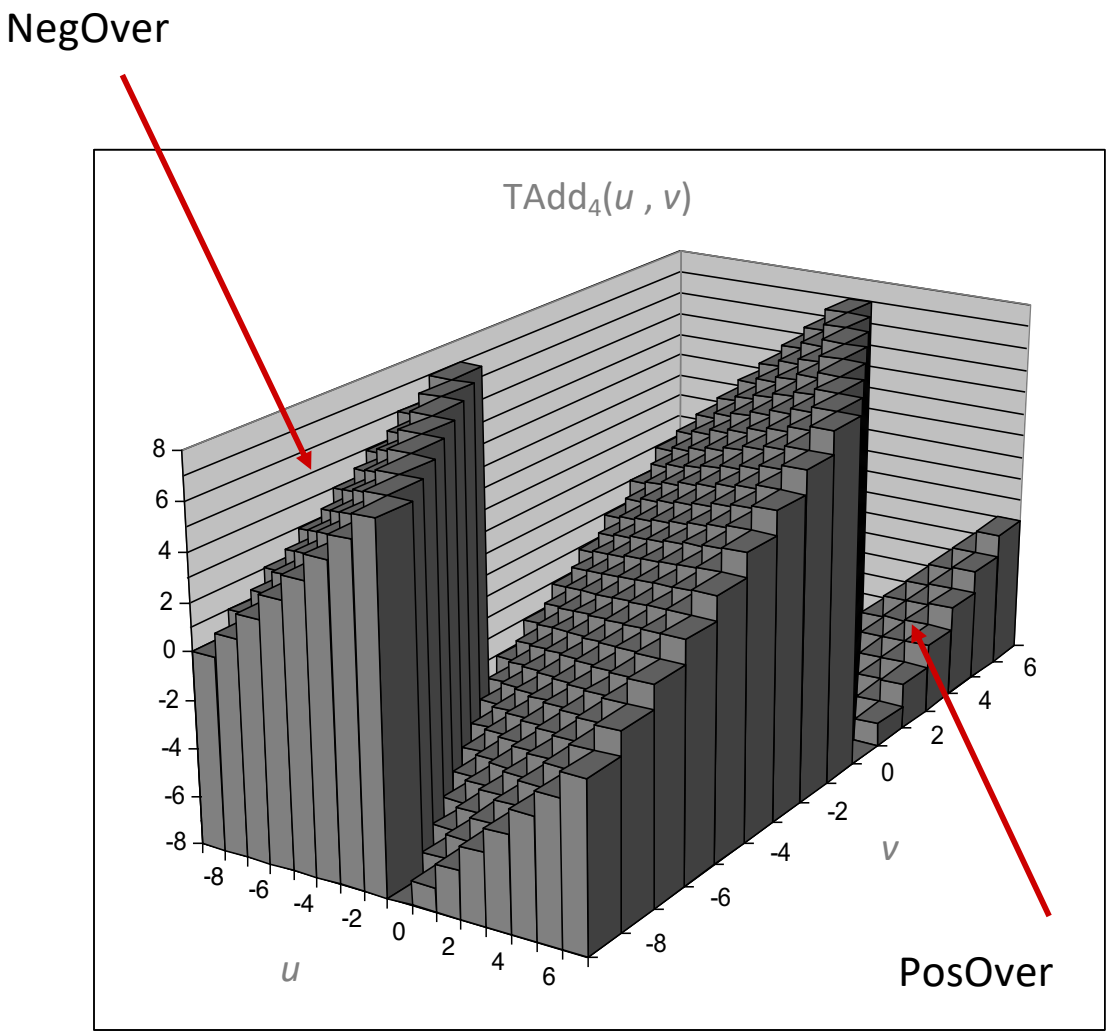
Becomes negative

At most once

If sum $< -2^{w-1}$

Becomes positive

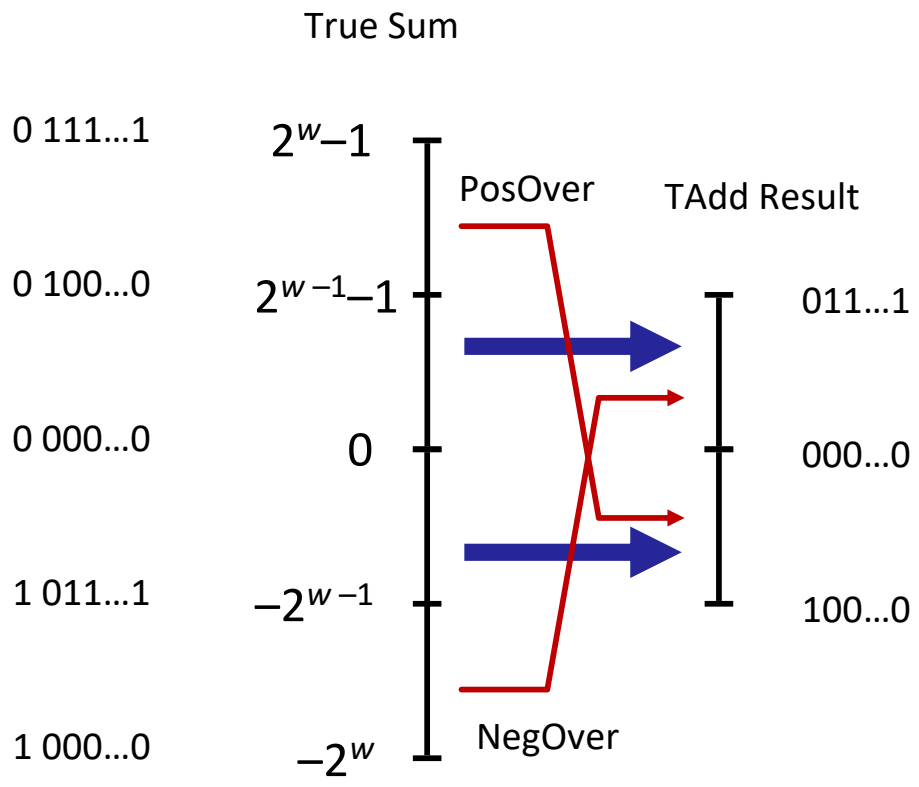
At most once



TAdd Overflow

Functionality

- True sum requires w+1 bits
- Drop off MSB
- Treat remaining bits as 2's comp. integer



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Shifting

Left Shift: `x << y`

- Shift bit-vector `x` left `y` positions
- Throw away extra bits on left
- Fill with 0's on right
- Equivalent to multiplying by 2^y

Right Shift: `x >> y`

- Shift bit-vector `x` right `y` positions
- Throw away extra bits on right
- Two kinds:
  - "Logical": Fill with 0's on left
  - "Arithmetic": Replicate most significant bit on left
- *Almost* equivalent to dividing by 2^y

Undefined Behavior (in C)

- Shift amount < 0 or ≥ word size

Argument `x`	`01100010`
<< 3	`00010000`
Logical >> 2	`00011000`
Arithmetic >> 2	`00011000`

Argument `x`	`10100010`
<< 3	`00010000`
Logical >> 2	`00101000`
Arithmetic >> 2	`11101000`

Multiplication

Goal: Computing Product of *w*-bit numbers *x*, *y*

Either signed or unsigned

But, exact results can be bigger than *w* bits

Unsigned: up to 2*w* bits

Result range: $0 \le x * y \le (2^w - 1)^2 = 2^{2w} - 2^{w+1} + 1$

Two's complement min (negative): Up to 2*w*-1 bits

Result range: $x * y \ge (-2^{w-1})*(2^{w-1}-1) = -2^{2w-2} + 2^{w-1}$

Two's complement max (positive): Up to 2*w* bits, but only for $(TMin_w)^2$

Result range: $x * y \le (-2^{w-1})^2 = 2^{2w-2}$

So, maintaining exact results…

would need to keep expanding word size with each product computed

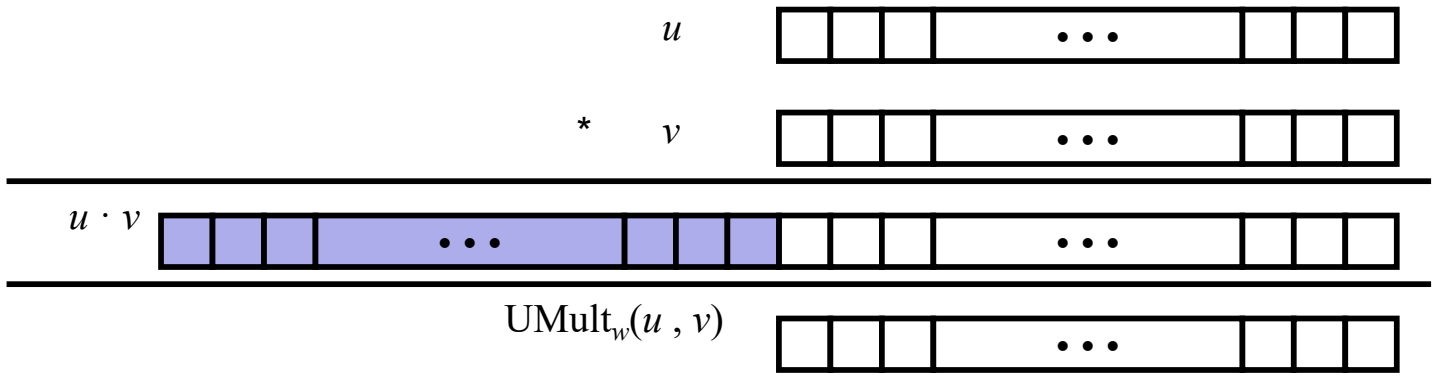
is done in software, if needed

Unsigned Multiplication in C

Operands: *w* bits

True Product: 2**w* bits

Discard *w* bits: *w* bits



Standard Multiplication Function

Ignores high order *w* bits

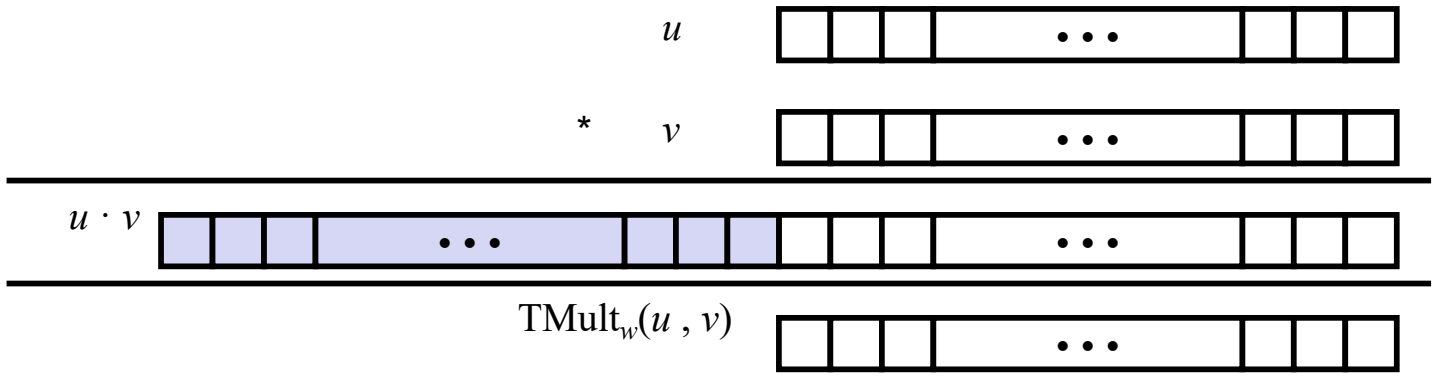
```
                1110 1001      E9        233
*               1101 0101   *  D5    *   213
 1100 0001 1101 1101         C1DD      49629
                1101 1101      DD        221
```

Signed Multiplication in C

Operands: *w* bits

True Product: 2**w* bits

Discard *w* bits: *w* bits



Standard Multiplication Function

Ignores high order *w* bits

Some of which are different for signed vs. unsigned multiplication

Lower bits are the same

```
                1110 1001      E9      -23
*               1101 0101   *  D5   *  -43
 0000 0011      1101 1101    03DD      989
                1101 1101      DD      -35
```

Power-of-2 Multiply with Shift

Operation

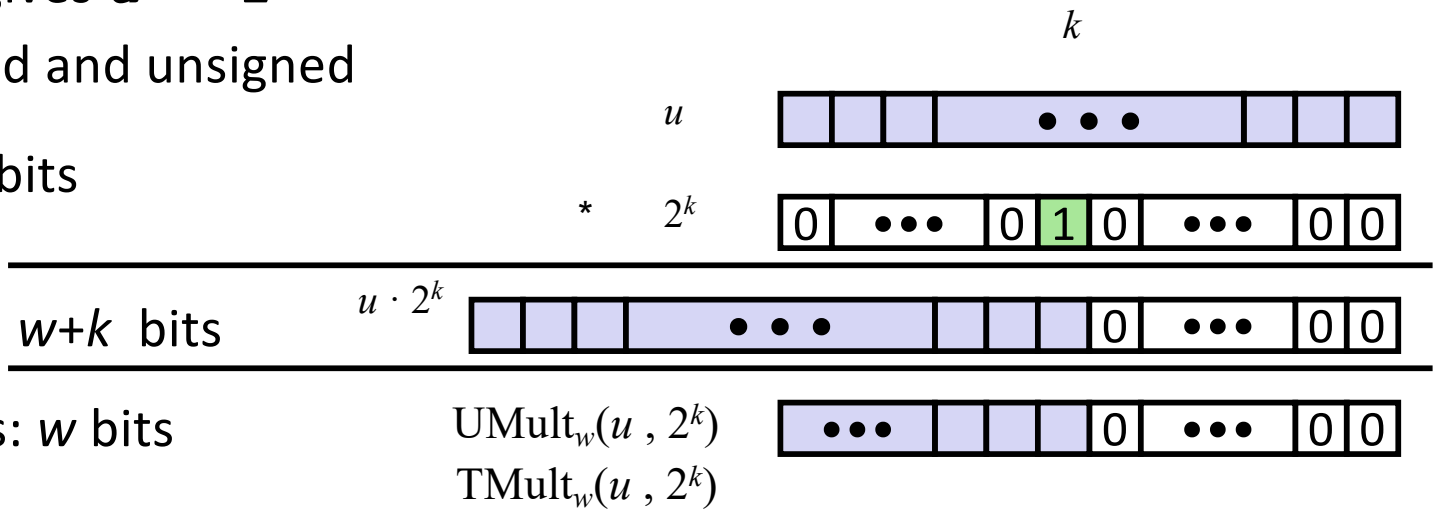
`u << k` gives `u * 2`^{k}

Both signed and unsigned

Operands: *w* bits

True Product: *w*+*k* bits

Discard *k* bits: *w* bits



Examples

```
u << 3          ==   u * 8
(u << 5) - (u << 3)      ==     u * 24
```

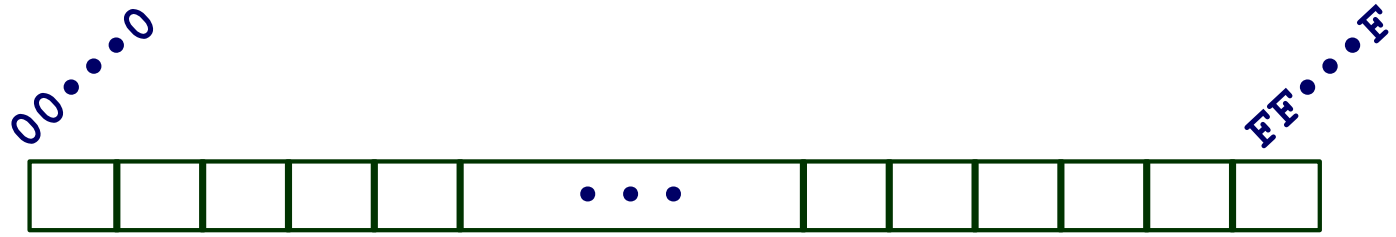
Most machines shift and add faster than multiply

Compiler generates this code automatically

Today: Bits, Bytes, and Integers

- Representing information as bits
- Bit-level manipulations
- Integers
  - Representation: unsigned and signed
  - Conversion, casting
  - Expanding, truncating
  - Addition, multiplication, shifting
- **Representations in memory, pointers, strings**

Byte-Oriented Memory Organization



Programs refer to data by address

- Imagine all of RAM as an enormous array of bytes
- An address is an index into that array
  - A pointer variable stores an address

System provides a private *address space* to each "process"

- A process is an instance of a program, being executed
- An address space is one of those enormous arrays of bytes
- Each program can see only its own code and data within its enormous array
- We'll come back to this later ("virtual memory" classes)

Machine Words

Any given computer has a "Word Size"

- Nominal size of integer-valued data
  - and of addresses

- Historically, most machines used 32 bits (4 bytes) as word size
  - Limits addresses to 4GB (2^{32} bytes)

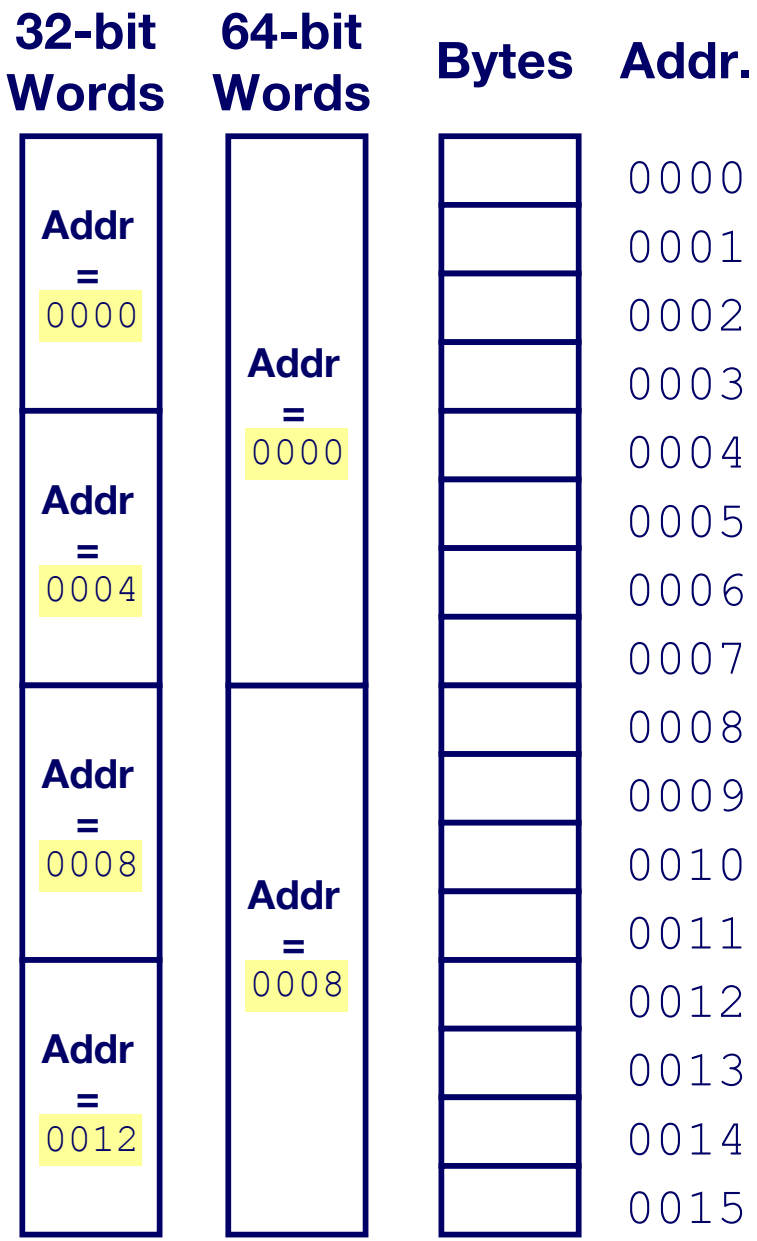
- Currently, machines have 64-bit word size
  - Potentially, could have 16 EB (exabytes) of addressable memory
  - That's 18.4×10^{18} bytes

- Machines still support multiple data formats
  - Fractions or multiples of word size
  - Always integral number of bytes

Addresses *Always* Specify Byte Locations

Address of a word is address of the first byte in the word

Addresses of successive words differ by 4 (32-bit) or 8 (64-bit)



Example Data Representations

C Data Type	Typical 32-bit	Typical 64-bit	x86-64
`char`	1	1	1
`short`	2	2	2
`int`	4	4	4
`long`	4	8	8
`float`	4	4	4
`double`	8	8	8
pointer	4	8	8

Byte Ordering

So, how are the bytes within a multi-byte word ordered in memory?

Conventions

- Big Endian: Sun, PPC Mac, *network packet headers*
  - Least significant byte has highest address
- Little Endian: *x86*, ARM processors running Android, iOS, and Windows
  - Least significant byte has lowest address

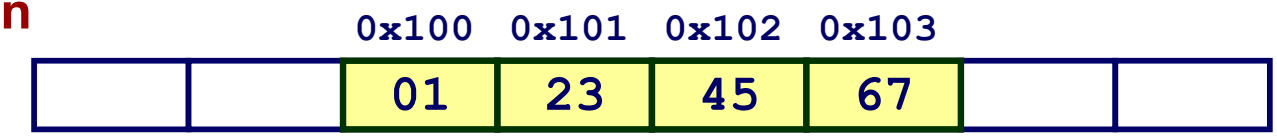
Byte Ordering Example

Example

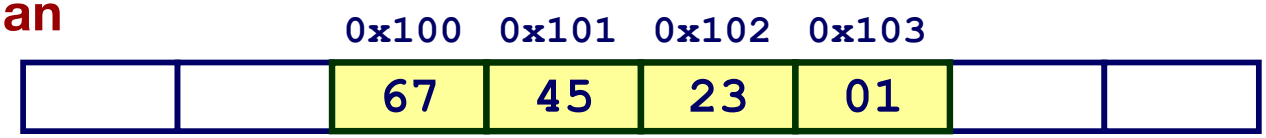
Variable x has 4-byte value of 0x01234567

Address given by &x is 0x100

Big Endian



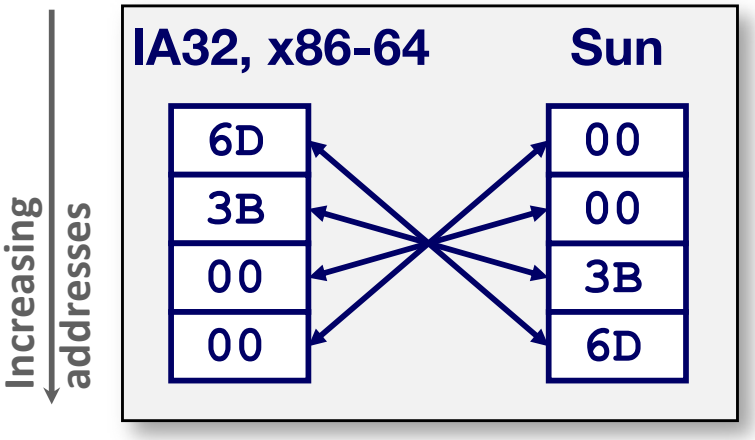
Little Endian



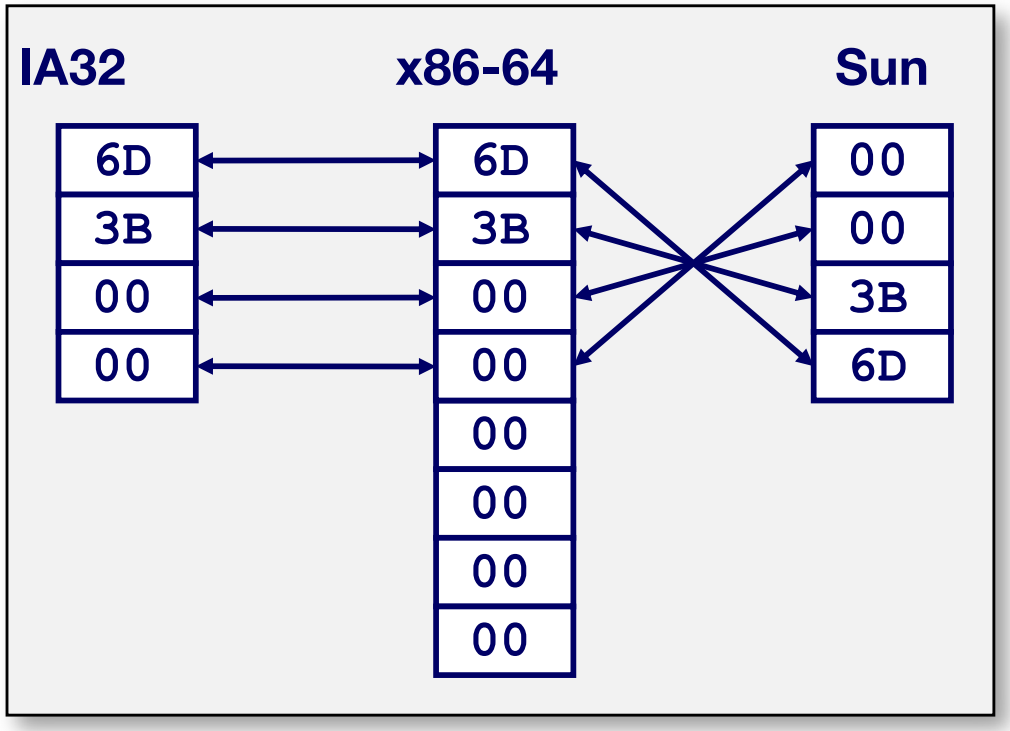
Representing Integers

Decimal:	15213
Binary:	0011 1011 0110 1101
Hex:	3 B 6 D

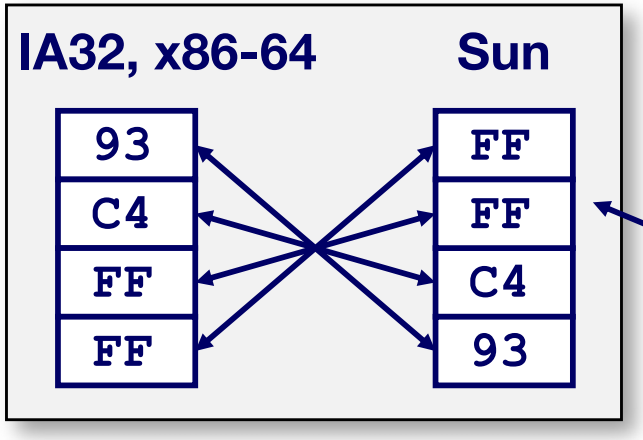
```
int A = 15213;
```




```
long int C = 15213;
```




```
int B = -15213;
```



Two's complement representation

Examining Data Representations

Code to Print Byte Representation of Data

Casting pointer to unsigned char * allows treatment as a byte array

```
typedef unsigned char *pointer;

void show_bytes(pointer start, size_t len){
  size_t i;
  for (i = 0; i < len; i++)
    printf("%p\t0x%.2x\n",start+i, start[i]);
  printf("\n");
}
```

Printf directives:

%p: Print pointer

%x: Print Hexadecimal

show_bytes Execution Example

```
int a = 985;
printf("int a = 985;\n");
show_bytes((pointer) &a, sizeof(int));
```

Result (Linux x86-64):

```
int a = 985;
0x7fffb7f71dbc    d9
0x7fffb7f71dbd    03
0x7fffb7f71dbe    00
0x7fffb7f71dbf    00
```

Representing Pointers

```
int B = -985;
int *P = &B;
```

Sun	IA32	x86-64
EF	AC	3C
FF	28	1B
FB	F5	FE
2C	FF	82
		FD
		7F
		00
		00

Different compilers & machines assign different locations to objects

Even get different results each time run program

Representing Strings

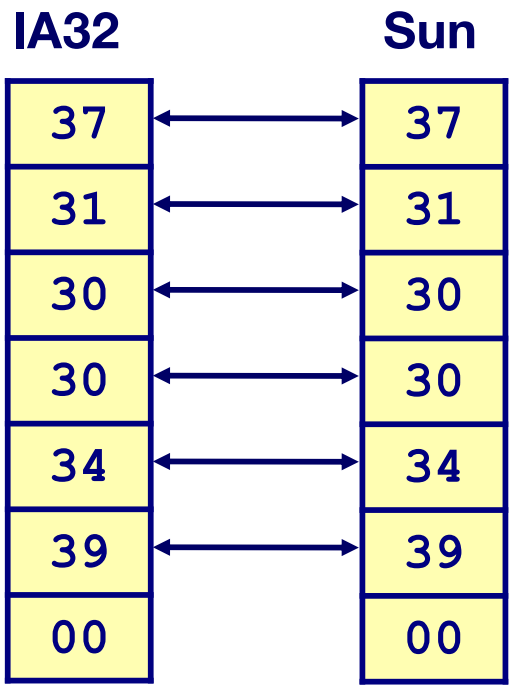
```
char S[7] = "710049";
```

Strings in C

- Represented by array of characters
- Each character encoded in ASCII format
  - Standard 7-bit encoding of character set
  - Character "0" has code 0x30
    - Digit *i* has code 0x30+*i*
- String should be null-terminated
  - Final character = 0

Compatibility

- Byte ordering not an issue



Representing x86 machine code

x86 machine code is a sequence of *bytes*

Grouped into variable-length instructions, which look like strings…

But they contain embedded little-endian numbers…

Example Fragment

Address	Instruction Code	Assembly Rendition
8048365:	5b	pop %ebx
8048366:	81 c3 ab 12 00 00	add $0x12ab,%ebx
804836c:	83 bb 28 00 00 00 00	cmpl $0x0,0x28(%ebx)

Deciphering Numbers

Value:	0x12ab
Pad to 32 bits:	0x000012ab
Split into bytes:	00 00 12 ab
Reverse:	ab 12 00 00